

Catalytic Impact of Early Investments in Advanced Clean Energy Technologies

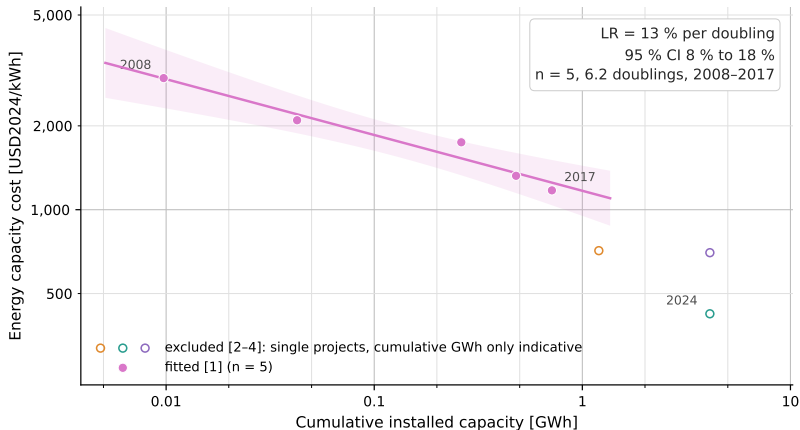
Technology cost

PyPSA Labs, Google

22 September 2026

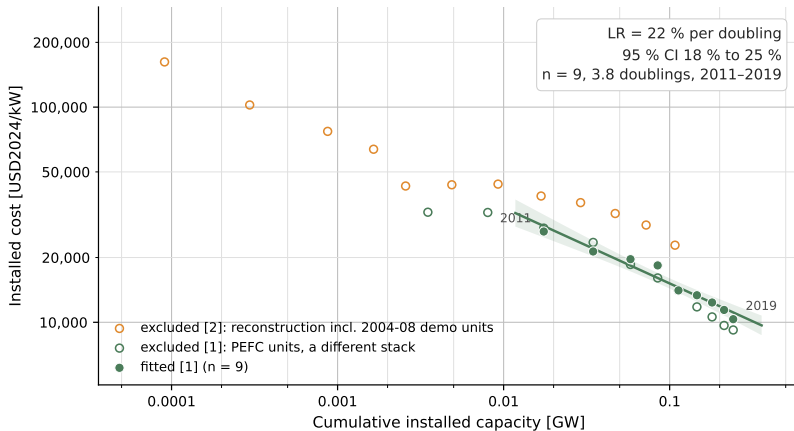


Vanadium flow battery: energy capacity cost



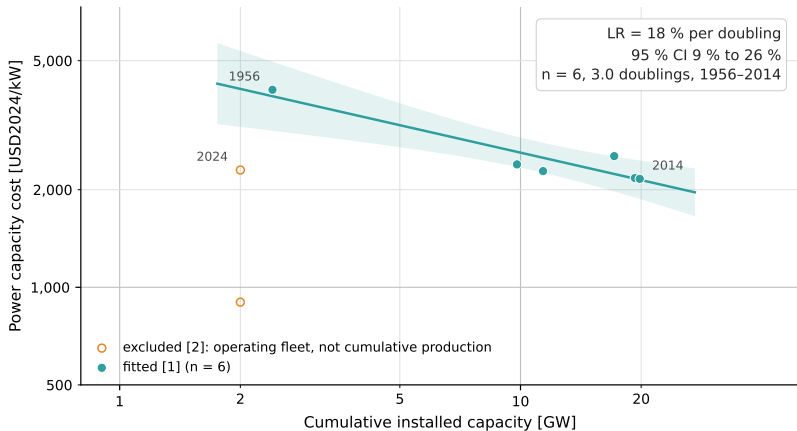
[1] Schmidt et al. (2017), Nat. Energy, 2018 update [2] Dalian Rongke 100 MW / 400 MWh [3] BNEF (2024) flow battery cost, China [4] BNEF (2024) flow battery cost, rest of world

Solid oxide fuel cell: ENE - FARM unit price



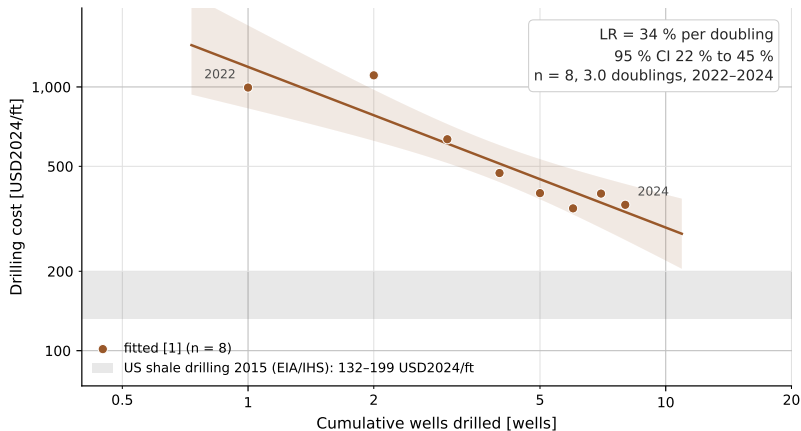
[1] ACE ENE - FARM shipment statistics (2025) with ANRE unit prices 2018 update [2] Schmidt et al. (2017), Nat. Energy,

(Alkaline) Electrolyser



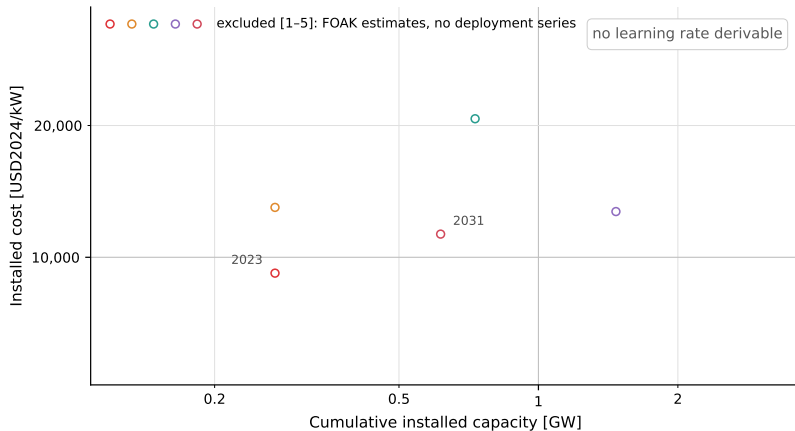
[1] Schmidt et al. (2017), Nat. Energy, 2018 update [2] IEA (2025), Global Hydrogen Review

Enhanced geothermal: Fervo drilling cost per foot



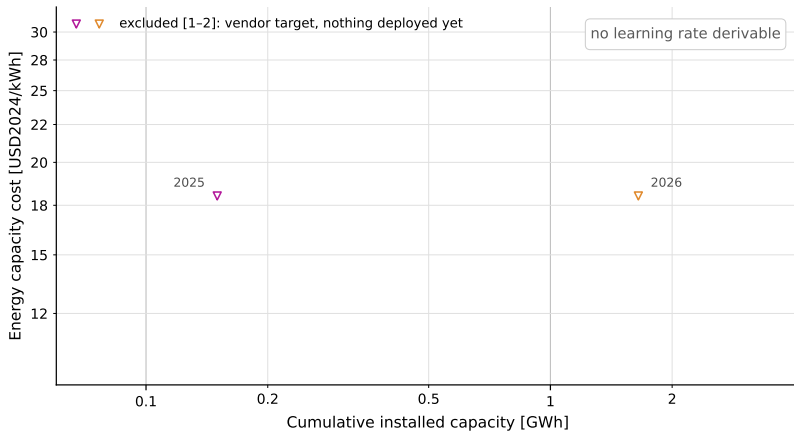
[1] El-Sadi et al. (2024) via Akindipe et al. (2025)

Nuclear SMR: FOAK estimates only



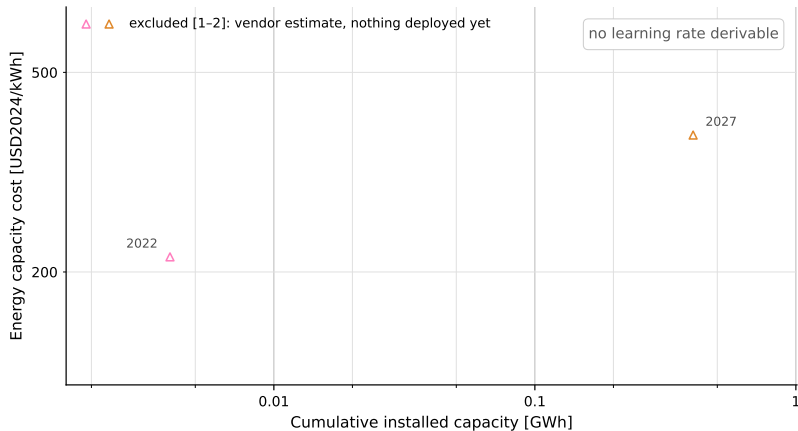
- [1] EIA AEO2025 [2] DOE Liftoff: Advanced Nuclear (2024) [3] NuScale / UAMPS CFPP estimate (2023)
[4] OPG Darlington BWRX-300 [5] TerraPower Natrium Kemmerer 1

Iron-air battery: vendor target only



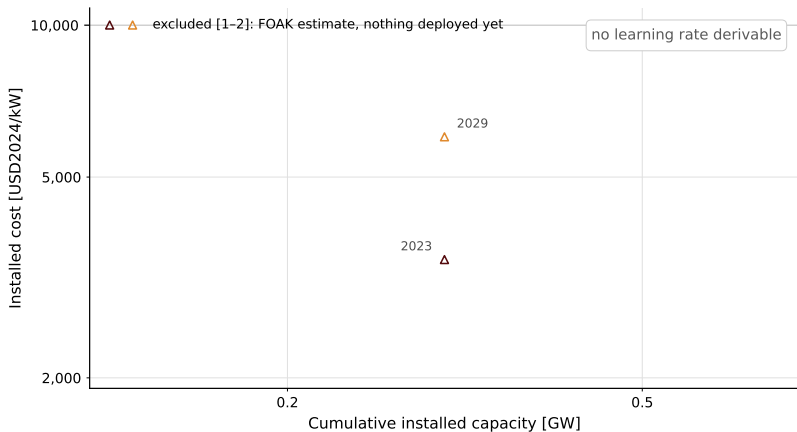
[1] Form Energy (2023), modelling recommendations [2] Form Energy (2023), vendor target

CO₂ battery: vendor estimates only



[1] Energy Dome, CEO estimate [2] Alliant Columbia Energy Storage (Energy Dome)

Allam - cycle gas + CCS: FOAK estimates only



[1] NET Power, Project Permian 2023 estimate [2] NET Power, Project Permian SN1 update

Closed-loop geothermal: FOAK estimate and projection only

